KENYA CERTIFICATE OF BASIC EDUCATION

JUNIOR SCHOOL EDUCATION ASSESSMENT

TERM 2: MIDTERM ASSESSMENT 2026

GRADE 9 – INTEGRATED SCIENCE PAPER 1

Time: 1 Hour 30 Minutes

LEARNER'S DETAILS

Name: _____. **School:** _____.

Assessment Number: _____. **Date:** _____.

School Code: _____. **Signature:** _____.

Instructions for Learners

1. Write your **full name** and **assessment number** in the spaces provided above.
2. Indicate your **school name and school code** clearly.
3. **Sign and date** this paper in the spaces provided.
4. This paper consists of **TWO sections**: Section A and Section B.
5. Answer **ALL questions in BOTH sections.**
6. Write your answers in the spaces provided in this question paper.
7. Do **NOT remove any page** from this paper.
8. All responses must be written in **English**

FOR OFFICIAL USE ONLY -FOR EXAMINER'S USE

SECTION	SECTION A	SECTION B	% SCORE	EE1	EE2	ME1	ME2	AE1	AE2	BE1	BE2
SCORE RANGE	20 MARKS	30 MARKS		90-100	75-89	58-74	41-57	31-40	21-30	11-20	1-10
POINTS				8 POINTS	7 POINTS	6 POINTS	5 POINTS	4 POINTS	3 POINTS	2 POINTS	1 POINT
LEARNER'S TOTAL SCORE											

SECTION A (20 MARKS)

Answer ALL questions in this section

1. During a class experiment on photosynthesis, learners tested a variegated leaf using iodine solution.
What observation would be made on the yellow part of the leaf?

A. Blue-black colour B. Yellow colour
C. Brown colour D. Green colour

2. In an environmental study lesson, learners classified components of the environment.
Which of the following is an abiotic component?

A. Plants B. Human being
C. Goat D. Salinity

3. While studying feeding relationships, learners discussed organisms that benefit each other.
Which type of nutrition describes this relationship?

A. Parasitism B. Symbiosis
C. Saprophytism D. Holozoic

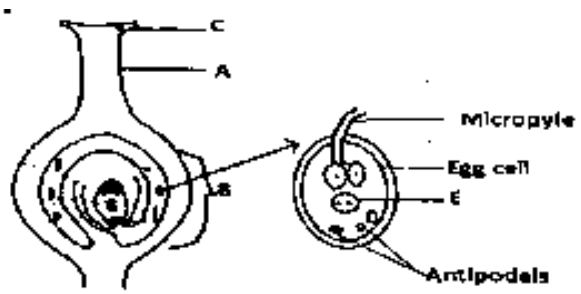
4. A farmer grows maize and beans on the same farm, and both compete for water and nutrients.
What type of competition is this?

A. Intraspecific B. Interspecific
C. Mutualism D. Commensalism

5. During a flower dissection practical, learners observed different parts of a flower.
Which part forms the carpel?

A. Anther B. Style C. Filament D. Petal

6. While studying fertilization in plants, learners observed a labelled diagram.
Which structure is labelled **E**?



A. Ovule B. Pollen grain
C. Polar nuclei D. Egg cell

7. In a biology lesson, learners studied a process occurring in flowering plants after pollination.
Which process was illustrated?

A. Pollination B. Fertilization
C. Germination D. Flowering

8. During the same lesson, learners identified parts of the flower.
What is the name of part A?

A. Style B. Stigma C. Ovary D. Pollen tube

9. In a food chain study, learners arranged organisms as follows:

Grass → Grasshopper → Mouse → Snake → Eagle.

What is the trophic level of the snake?

A. Tertiary consumer B. Primary consumer
C. Producer D. Quaternary consumer

10. Learners studied seed dispersal by animals in class.
Which of the following is NOT a characteristic of such fruits?

A. Succulent B. Scented
C. Brightly coloured D. Parachute

11. In a chemistry practical, copper (II) sulphate crystals were dissolved in water.
What colour was observed?

A. Blue B. Green C. Colourless D. Pink

12. During oil extraction experiments, learners used octane instead of water.
Why was octane preferred?

A. It dissolves oil better B. It is safer
C. It reacts with seeds D. It changes oil colour

13. While identifying acids and bases at home, learners tested different substances.
Which one is basic?

A. Soap B. Vinegar
C. Lemon juice D. Tomato sauce

14. In a lesson on excretion, learners discussed substances removed by the kidneys.
Which substance is NOT excreted by the kidney?

A. Urea B. Water C. Oxygen D. Salt

15. A learner used salt to kill snails and observed that they lost water and died.
Which process caused this?

A. Dehydration B. Diffusion
C. Active transport D. Osmosis

16. While repairing an appliance, a technician found that current was not flowing due to a break.
What is this condition called?

A. Closed circuit B. Open circuit
C. Short circuit D. Parallel circuit

17. In a chemistry lesson, an element with electronic configuration 2.8.7 was discussed.
Which element is this?

A. Oxygen B. Fluorine C. Argon D. Chlorine

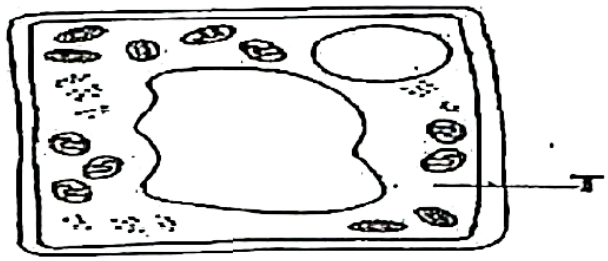
18. During atomic structure calculations, an atom had mass number 28 and atomic number 14.
How many neutrons does it have?

A. 14 B. 28 C. 15 D. 13

19. In a physics experiment, learners applied equal force on different surfaces.
Which surface produced the highest pressure?

A. Large flat surface B. Wide tyre
C. Small pointed surface D. Wooden plank

20. While observing a plant cell under a microscope, learners identified labelled parts.



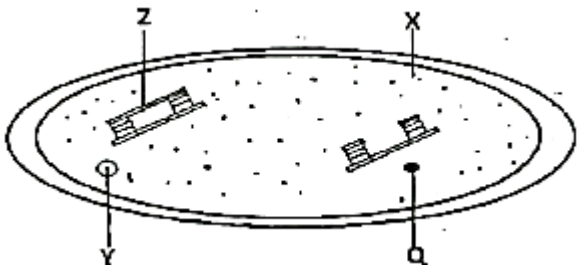
Which structure is labelled T?

A. Vacuole B. Nucleus
C. Cytoplasm D. Cell membrane

21. In a renewable energy lesson, learners studied wind turbines.
What energy conversion occurs in a wind turbine?

A. Kinetic to light B. Kinetic to electrical
C. Electrical to kinetic D. Chemical to wind

22. During a chloroplast study, learners observed internal structures.



Which labelled part represents the lamella?

A. Y B. Z C. X D. O

23. A fire broke out in a computer lab due to an electrical fault.
Which extinguisher should be used?

A. Dry powder B. Foam
C. Water D. Wet chemical

24. In a laboratory experiment, a learner needed exactly 25 ml of liquid.
Which apparatus is most suitable?

A. Pipette
B. Burette
C. Measuring cylinder
D. Conical flask

25. During a health lesson, learners discussed ways of preventing STIs.
Which method is correct?

A. Avoid crowded places
B. Unprotected sex
C. Abstinence
D. Avoid vaccination

26. While studying plant cells, learners identified storage structures.
Which part stores sap and waste?

A. Nucleus
B. Chloroplast
C. Vacuole
D. Cytoplasm

27. In a health education lesson, learners studied sexually transmitted infections.
Which STI is caused by a virus and leads to weight loss?

A. Gonorrhoea
B. Syphilis
C. Herpes
D. HIV/AIDS

28. In a chemistry lesson, learners studied element symbols derived from Latin.
Which element has the symbol Fe?

A. Silver
B. Iron
C. Copper
D. Lead

29. During fire safety training, learners classified types of fires.
Which class involves electrical appliances?

A. Class A
B. Class C
C. Class E
D. Class F

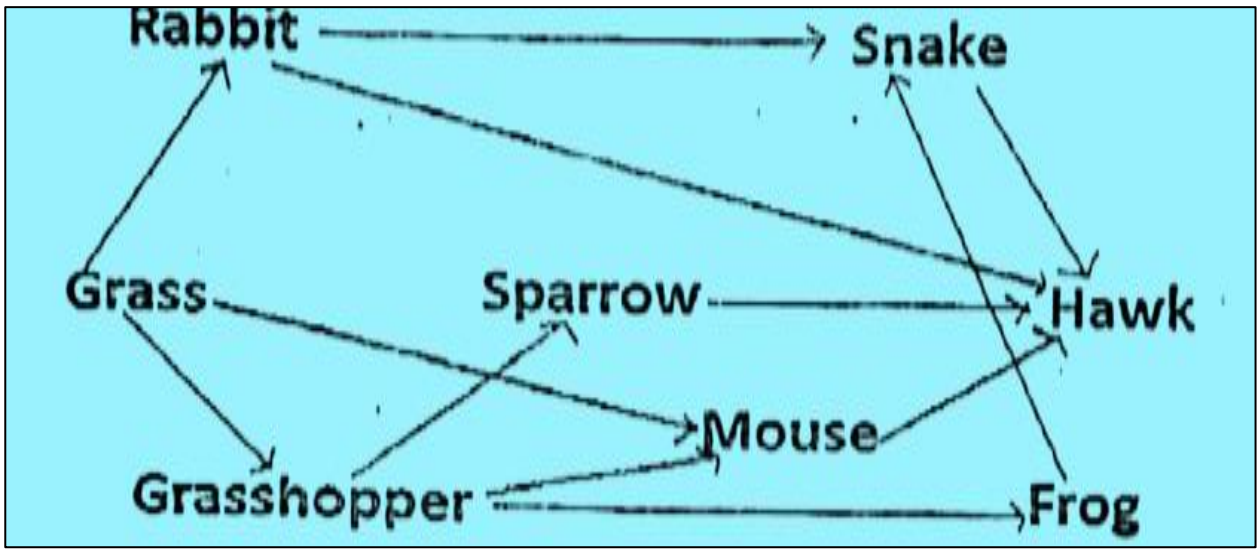
30. In a boiling point experiment, learners compared different solutions.
Which solution has the highest boiling point?

A. Pure water
B. Distilled water
C. Salt water
D. Rain water

SECTION B (40 MARKS)

Answer ALL questions in this section

31. During an ecology lesson, learners studied a food web involving grass, rabbit, grasshopper, frog, snake, hawk, sparrow and mouse.



a) Identify the organism at the highest trophic level. (1 mk)

_____.

b) Construct a food chain ending with a hawk as a tertiary consumer. (2 mks)

_____

_____

_____

c) Identify the producer. (1 mk)

_____.

d) State the trophic level of:

(i) Grasshopper _____.(1 mk)

(ii) Frog _____. (1 mk)

e) State the ultimate source of energy. (1 mk)

_____.

f) Explain what would happen if all snakes were removed. (2 mks)

_____

_____

32. In a food chain study, learners discussed energy transfer.
State two ways energy is lost. (2 mks)

i. _____.
ii._____.

33. In a physics experiment, a box measuring 0.4 m × 0.2 m exerts a force of 80 N on a table.

a) Calculate the pressure exerted. (3 mks)

b) Explain why tractors have wide tyres. (2 mks)

_____

_____

34. In an energy lesson, learners studied different devices.
State the energy transformations in:

a) Radio (1 mk)

_____

_____

b) Solar panel (1 mk)

_____

_____

35. In a biology lesson, learners studied diffusion in animals.
State two roles of diffusion. (2 mks)

i. _____.
ii._____.

36. During digestion studies, learners examined gastric juice.

a) List three components of gastric juice. (3 mks)

i. _____.
ii._____.
iii. _____.

b) State the role of sodium hydrogen carbonate in the duodenum. (1 mk)

_____

_____

37. In a chemistry experiment, learners tested different water samples using soap solution.

Water sample	**Volume of soap required before boiling (cm^3)**	**Volume of soap required after boiling (cm^3)**
X	8.0	4.0
Y	12.0	6.0
Z	2.5	2.5
V	7.0	3.0

a) State what should be kept constant for fair comparison. (1 mk)

_____

_____

b) Identify samples with temporary hardness. (2 mks)

_____

_____

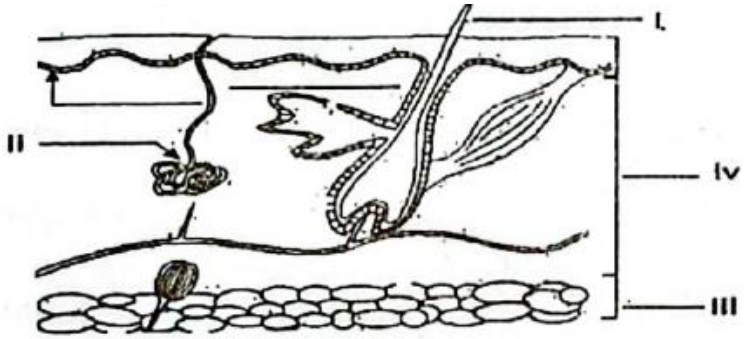
c) Identify the distilled water sample. (1 mk)

_____

_____

38. During a biology lesson, learners studied the structure of mammalian skin.

Use the diagram provided to answer the questions. Name the parts labeled:



i. _____.

ii. _____.

iii. _____.

iv. _____.

39. In a plant reproduction lesson, learners studied wind-pollinated flowers. State three characteristics. (3 mks)

i. _____.
ii._____.
iii. _____.

40. In a materials science lesson, learners studied alloys.
Brass is made of which two metals? (2 mks)

i. _____.
ii._____.

41. While in a forest, a learner observed a decaying antelope carcass.

a) Name two decomposers involved. (2 mks)

i. _____.
ii._____.

b) State their role. (1 mk)

_____

_____

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MARKING SCHEME

SECTION A: MULTIPLE CHOICE (20 MARKS)

(1 mark each)

Qn	Answer
1	B
2	D
3	B
4	B
5	B
6	C
7	B
8	B
9	A
10	D
11	A
12	A
13	A
14	C
15	D
16	B
17	D
18	A
19	C
20	C
21	B
22	B
23	A
24	A
25	C
26	C
27	D
28	B
29	C
30	C

SECTION B: STRUCTURED QUESTIONS (40 MARKS)

Q31: FOOD WEB (8 MARKS)

a) Highest trophic level (1 mk)

✔ Hawk

b) Food chain (2 mks)

✔ Example:
Grass → Grasshopper → Frog → Snake → Hawk
OR
Grass → Rabbit → Snake → Hawk

✔ Must end with hawk as tertiary/quaternary consumer

c) Producer (1 mk)

✔ Grass

d) Trophic levels (2 mks)

i) Grasshopper → Primary consumer
ii) Frog → Secondary consumer

e) Source of energy (1 mk)

✔ Sun

f) Effect of removing snakes (2 mks)

✔ Expected points:

- Increase in prey population (frogs/rabbits)
- Disturbance of food chain balance
- Increase in grass destruction
- Decrease in hawk food supply

Q32: ENERGY LOSS (2 MARKS)

✔ Any two:

- Heat loss to environment
- Movement/respiration
- Waste (faeces, urine)
- Incomplete digestion

Q33: PRESSURE (5 MARKS)

a) Calculation (3 mks)

Area = $0.4 \times 0.2 = 0.08\ m^2$

Pressure = Force / Area
= 80 / 0.08
= **1000 Pa**

✔ Final answer: **1000 Pascal**

b) Tractors wide tyres (2 mks)

✔ Keywords:

- Increase surface area
- Reduce pressure on soil
- Prevent sinking

Q34: ENERGY TRANSFORMATION (2 MARKS)

a) Radio
✔ Electrical → Sound energy

b) Solar panel
✔ Light → Electrical energy

Q35: DIFFUSION (2 MARKS)

✔ Any two:

- Gas exchange in lungs
- Movement of oxygen in blood
- Removal of CO_2
- Absorption in alveoli

Q36: DIGESTION (4 MARKS)

a) Gastric juice components (3 mks)

✔ Any three:

- Hydrochloric acid
- Pepsin
- Mucus
- Water

b) Sodium hydrogen carbonate role (1 mk)

✔ Neutralizes acidic chyme
✔ Provides alkaline medium in duodenum

Q37: HARDNESS OF WATER (4 MARKS)

a) Constant factor (1 mk)

✔ Same volume of water sample
OR
✔ Same soap concentration

b) Temporary hardness (2 mks)

✔ Samples:

- X
- Y
- V

(Decrease after boiling)

c) Distilled water (1 mk)

✔ Z (no change before and after boiling)

Q38: MAMMALIAN SKIN (DIAGRAM QUESTION)

i. Hair
ii. Sweat gland
iii. Malphigian layer
iv. Dermis

Q39: WIND POLLINATION (3 MARKS)

✔ Any three:

- Small, light pollen grains
- Feathery stigma
- Large anthers exposed
- No scent or bright colours
- Long filaments
- Lots of pollen produced

Q40: BRASS (2 MARKS)

✔ Copper + Zinc

Q41: DECOMPOSERS (3 MARKS)

a) Two decomposers (2 mks)

✔ Any two:

- Bacteria
- Fungi

b) Role (1 mk)

✔ Break down dead organic matter
✔ Release nutrients into soil
✔ Recycling of nutrients